CS1010 Discrete Mathematics for Computer Science

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Poisson Distribution

The Poisson distribution models the number of times an event occurs during a fixed interval of time, distance, area, etc.

Examples:

- Number of calls received by a call center in one minute.
- Number of defects in one meter of manufactured cable.
- Number of α-particles detected in one second.
- Number of requests received by a server in one minute.

If X is Poisson with parameter $\lambda > 0$, then

$$\boxed{\Pr(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}} \text{ where } k = 0, 1, 2, \ldots.$$



Interpreting the Poisson Parameter

For a Poisson process, λ represents the **average rate of occurrence** over the interval being considered. For example, suppose a server receives

3.2 requests per second.

- For a one-second interval, we use $\lambda = 3.2$.
- For a two-second interval, $\lambda = 6.4$.
- For a 10-second interval, $\lambda = 32$.

Thus, the parameter changes when the size of the interval changes.



Application of the Poisson Distribution

Suppose a server receives

$$3.2 \text{ requests per second.}$$

Let X denote the number of requests in one second.

$\mathcal{Q}$. What is the probability that at most two requests are received?

$\mathcal{A}$. $\Pr(X \leq 2) = \sum_{k=0}^{2} e^{-3.2} \frac{3.2^k}{k!}$.

Binomial vs. Poisson

Both distributions count the number of occurrences of an event, but they arise in different settings.

	Binomial	**Poisson**
What is fixed?	Number of trials n	Size of interval
What is counted?	Number of successes	Number of occurrences
Parameters	n, p	λ

Example:

- Number of defective items among 100 tested items $\rightarrow$ Binomial.
- Number of defects in one meter of material $\rightarrow$ Poisson.



Poisson Approximation to the Binomial

Suppose $X \sim \text{Binomial}(n, p)$.

When n is large and p is small, the binomial distribution can be approximated by a Poisson distribution with

$$\boxed{\lambda = np}.$$

Thus,

$$\binom{n}{k} p^k (1-p)^{n-k} \approx e^{-np} \frac{(np)^k}{k!}.$$

Intuition: Many trials + very small probability of success $\rightarrow$ Poisson distribution.

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Q&A: Which Distribution?

Identify the most appropriate distribution.

1. A fair coin is tossed once. Let $X = 1$ if the result is heads and $X = 0$ otherwise.



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2. A fair coin is tossed 20 times. Let X be the number of heads.



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3. A fair coin is tossed until the first head appears. Let X be the number of tosses.



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4. A server receives requests at a rate of 5 requests per second. Let X be the number of requests received during one second.



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   Poisson



Discrete Uniform Distribution

Suppose a random variable X can take n possible values, and every value is equally likely. Then X has a **discrete uniform distribution**:

$$\Pr(X = x) = \frac{1}{n}.$$

Examples:

- Outcome of a fair die.
- Result of a uniformly chosen card from a deck.
- A uniformly chosen integer from $\{1, \ldots, n\}$.

For a fair six-sided die,

$$\Pr(X = 1) = \cdots = \Pr(X = 6) = \frac{1}{6}.$$



Choosing a Probability Distribution

Situation	Distribution
One success/failure experiment	Bernoulli
Number of successes in n trials	Binomial
Number of trials until first success	Geometric
Number of events in a fixed interval	Poisson
All possible outcomes equally likely	Uniform

The key is to identify what is being counted.



Q&A: Biased Coin

$\mathcal{Q}$. A biased coin satisfies

$$\Pr(H) = p, \qquad \Pr(T) = 1 - p.$$

The coin is tossed n times. What is the probability that the first head occurs exactly on the k-th toss?

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$\mathcal{A}$. The first $k - 1$ tosses must be tails and the k-th toss must be a head. Hence

$$\boxed{\Pr(\text{first head on toss } k) = (1 - p)^{k-1} p}.$$

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$\mathcal{A}$. Let X be the number of heads. Then

$$X \sim \text{Binomial}(n, p).$$

Therefore,

$$\Pr(X \geq k) = \sum_{j=k}^{n} \binom{n}{j} p^j (1-p)^{n-j}.$$

Equivalently,

$$\boxed{\Pr(X \geq k) = 1 - \sum_{j=0}^{k-1} \binom{n}{j} p^j (1-p)^{n-j}}.$$

Q&A: Two Biased Coins

$\mathcal{Q}$. Coin A has $\Pr(H) = p$ and coin B has $\Pr(H) = q$. One coin is chosen uniformly at random and tossed twice. What is the probability of obtaining exactly one head?

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$\mathcal{A}$. Condition on which coin was selected:

$$\begin{aligned}\Pr(\text{exactly one } H) &= \frac{1}{2}[2p(1-p)] + \frac{1}{2}[2q(1-q)] \\ &= p(1-p) + q(1-q).\end{aligned}$$

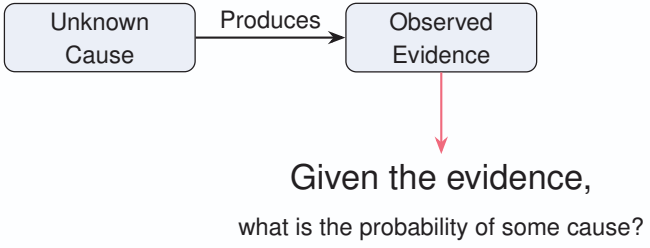
Thus,

$$\boxed{\Pr(\text{exactly one } H) = p(1-p) + q(1-q)}.$$

Bayes' Theorem

Motivation

Often, we observe some **evidence** and want to infer its most likely **cause**:



- A medical test is positive. What is the probability that the patient actually has the disease?
- An email is marked as spam. What is the probability that it really is spam?

Introduction

Bayes' theorem provides a systematic way to **update probabilities when new evidence is observed**.

Bayes' Theorem

Suppose E and F are events from a sample space S such that $\Pr(E) \neq 0$ and $\Pr(F) \neq 0$. Then:

$$\Pr(F|E) = \frac{\Pr(E|F)\Pr(F)}{\Pr(E|F)\Pr(F) + \Pr(E|\overline{F})\Pr(\overline{F})}$$

Bayes' Theorem Example

Q. Consider the following scenario:

- 20% of incoming emails are spam.
- The spam filter correctly flags spam emails with probability 0.98.
- It incorrectly flags a legitimate email as spam with probability 0.04.

If an email is marked as spam, what is the probability that it is actually spam?



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- 20% of incoming emails are spam.
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If an email is marked as spam, what is the probability that it is actually spam?

$\mathcal{A}$. Let $S = \texttt{email is spam}$, $F = \texttt{email is flagged as spam}$. Then we have

$$\Pr(S) = 0.2, \quad \Pr(F|S) = 0.98, \quad \Pr(F|\overline{S}) = 0.04.$$

Therefore,

$$\Pr(S|F) = \frac{0.98(0.2)}{0.98(0.2) + 0.04(0.8)} \approx 0.86.$$

Bayes' Theorem Example

Q. Box 1: 2 green, 7 red balls. Box 2: 4 green, 3 red balls. If you selects a box at random, and then a ball. If the ball is red, what is the probability it came from Box 1?

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$\mathcal{A}$. Let E be choosing a red ball, F be choosing Box 1.

$$\Pr(F) = 1/2, \quad \Pr(\overline{F}) = 1/2$$

$$\Pr(E|F) = 7/9, \quad \Pr(E|\overline{F}) = 3/7$$

$$\Pr(F|E) = \frac{(7/9)(1/2)}{(7/9)(1/2) + (3/7)(1/2)} = \frac{7/9}{7/9 + 3/7} = \frac{49}{76} \approx 0.645$$

Derivation of Bayes' Theorem

$$\Pr(F|E) = \frac{\Pr(E|F)\Pr(F)}{\Pr(E|F)\Pr(F) + \Pr(E|\overline{F})\Pr(\overline{F})}$$

Derivation of Bayes' Theorem

From the definition of conditional probability:

$$\Pr(E \cap F) = \Pr(E|F)\Pr(F) \quad \text{and} \quad \Pr(E \cap F) = \Pr(F|E)\Pr(E)$$

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Since $\Pr(E) = \Pr(E|F)\Pr(F) + \Pr(E|\overline{F})\Pr(\overline{F})$, we get:

$$\Pr(F|E) = \frac{\Pr(E|F)\Pr(F)}{\Pr(E|F)\Pr(F) + \Pr(E|\overline{F})\Pr(\overline{F})}$$

Applying Bayes' Theorem: Medical Testing

Example

One person in 100,000 has a disease. The test is 99% accurate for sick people ($\Pr(\text{Pos}|\text{Sick}) = 0.99$) and has a false positive rate of 0.5% ($\Pr(\text{Pos}|\text{Healthy}) = 0.005$). Find:

a) The probability a person testing positive actually has the disease.

b) The probability a person testing negative does not have the disease.

Medical Testing Solution

Let D be having the disease, E be testing positive.

$$\Pr(D) = 0.00001, \quad \Pr(\overline{D}) = 0.99999$$

$$\Pr(E|D) = 0.99, \quad \Pr(E|\overline{D}) = 0.005$$

$$\Pr(D|E) = \frac{(0.99)(0.00001)}{(0.99)(0.00001) + (0.005)(0.99999)} \approx 0.00198$$

So, only about 0.2% of people testing positive actually have the disease!

Medical Testing Solution (Negative Result)

What if the result is negative ($\overline{E}$)?

$$\Pr(\overline{D}|\overline{E}) = \frac{\Pr(\overline{E}|\overline{D})\Pr(\overline{D})}{\Pr(\overline{E}|\overline{D})\Pr(\overline{D}) + \Pr(\overline{E}|D)\Pr(D)}$$

$$\Pr(\overline{E}|\overline{D}) = 0.995, \quad \Pr(\overline{E}|D) = 0.01$$

$$\Pr(\overline{D}|\overline{E}) = \frac{(0.995)(0.99999)}{(0.995)(0.99999) + (0.01)(0.00001)} \approx 0.9999999$$

It is extremely unlikely to have the disease if you test negative.



Q&A Bayes' Theorem

Question. An LLM generates the token "Paris". Suppose that 60% of the time the model's context is about **travel** and 40% about **history**. The token "Paris" occurs with probability 0.8 in travel contexts and 0.3 in history contexts.

What is the probability that the context is about travel, given that the model generated "Paris"?

$\mathcal{Q}$. Let $T = \{\text{travel context}\}$ and $P = \{\text{model generates "Paris"}\}$.

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What is the probability that the context is about travel, given that the model generated "Paris"?

$\mathcal{Q}$. Let $T = \{\text{travel context}\}$ and $P = \{\text{model generates "Paris"}\}$.

$\mathcal{A}$. By Bayes' theorem, $\Pr(T \mid P) = \frac{\Pr(P|T)\Pr(T)}{\Pr(P|T)\Pr(T)+\Pr(P|\overline{T})\Pr(\overline{T})}$.

$\mathcal{A}$. Substituting the values, $\Pr(T \mid P) = \frac{(0.8)(0.6)}{(0.8)(0.6)+(0.3)(0.4)} = \frac{0.48}{0.60} = \boxed{0.8}$.

Generalized Bayes' Theorem

Theorem

Suppose E is an event from S and $F_1, F_2, \ldots, F_n$ are mutually exclusive events such that $\bigcup_{i=1}^{n} F_i = S$. Assume $\Pr(E) \neq 0$ and $\Pr(F_i) \neq 0$. Then:

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$\Pr(F_j|E) = \frac{\Pr(E|F_j)\Pr(F_j)}{\Pr(E)}$

$\Pr(E) = \sum_{i=1}^{n} \Pr(E|F_i)\Pr(F_i)$.

Substituting the latter into the former gives the result.



Monty Hall Problem

Suppose you choose door i and the winning door W is uniformly distributed over $1, 2, 3$. Let M be the door opened by Monty.

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$\mathcal{A}$. Since Monty never opens the chosen or winning door,

$$\Pr(M = j \mid W = k) = \begin{cases} \frac{1}{2} \text{ if } j \neq i, i = k, \\ 1, \text{ if } j, i, k \text{ all distinct} \\ 0, \text{ if } j = i \text{ or } j = k \end{cases}$$

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Bayes' Theorem

Bayesian Spam Filters

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A **spam filter** tries to automatically determine whether an incoming message is spam.

- One approach is to use **Bayes' theorem** together with information from previously classified messages.
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- One approach is to use **Bayes' theorem** together with information from previously classified messages.
- A **Bayesian spam filter** looks for words that provide evidence about whether a message is spam.
- "offer", "special", "opportunity" $\rightarrow$ may indicate spam
- "mom", "lunch", "Jan" $\rightarrow$ may indicate non-spam



Bayesian Spam Filters: False Positives

Suppose a spam filter classifies every incoming message as either **spam** or **not spam**.

	Actually Spam	**Actually Not Spam**
Classified as Spam	Correct	**False Positive**
Classified as Not Spam	**False Negative**	Correct

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Classified as Not Spam	**False Negative**	Correct

In practice, false positives are particularly undesirable: losing an important legitimate e-mail can be worse than allowing some spam through.

Learning from Previously Classified Messages

Suppose we have two collections of previously classified messages:

1. B: a set of messages known to be **spam** (“bad” messages).
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For a particular word w, let

$$n_B(w) = \text{number of spam messages containing } w,$$

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We estimate:

$$p(w) = \frac{n_B(w)}{|B|} \qquad \text{and} \qquad q(w) = \frac{n_G(w)}{|G|}.$$



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- $p(w)$: estimated probability that a **spam** message contains w.
- $q(w)$: estimated probability that a **non-spam** message contains w.

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- Assuming that an incoming message is equally likely to be spam or non-spam, $\Pr(S) = \Pr(\overline{S}) = \frac{1}{2}$.
- Therefore,

$$\Pr(S \mid E) = \frac{p(w)}{p(w) + q(w)}$$

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- **Decision rule:** If this probability exceeds a chosen threshold (e.g., 0.9), classify the message as **spam**.

Bayesian Spam Filter: A Single Word

The word “Rolex” occurs in 250 out of 2000 spam messages, and 5 out of 1000 non-spam messages. If the threshold is 0.9, should we classify the message as spam?



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$\mathcal{A}$. The probability of “Rolex” in spam is $\Pr(\text{Rolex}) = \frac{250}{2000} = 0.125$.



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$\mathcal{A}$. The probability of “Rolex” in non-spam is $q(\text{Rolex}) = \frac{5}{1000} = 0.005$.



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The word “Rolex” occurs in 250 out of 2000 spam messages, and 5 out of 1000 non-spam messages. If the threshold is 0.9, should we classify the message as spam? **Solution**

$\mathcal{A}$. The probability of “Rolex” in spam is $\Pr(\text{Rolex}) = \frac{250}{2000} = 0.125$.

$\mathcal{A}$. The probability of “Rolex” in non-spam is $q(\text{Rolex}) = \frac{5}{1000} = 0.005$.

$\mathcal{Q}$. The spam score is

$$r(\text{Rolex}) = \frac{p(\text{Rolex})}{p(\text{Rolex}) + q(\text{Rolex})} = \frac{0.125}{0.125 + 0.005} \approx 0.9615.$$

$\mathcal{A}$. Since $0.9615 > 0.9$, we classify the message as **spam** and reject it.

Bayes' Theorem: Spam Detection

Question. Let E_1 and E_2 be the events that an incoming message contains words w_1 and w_2, respectively.

Assume E_1, E_2 are independent, both conditional on S (spam) and on $\overline{S}$ (not spam). With no prior knowledge, $\Pr(S) = \Pr(\overline{S}) = \frac{1}{2}$.

$\mathcal{Q}$. Find $\Pr(S \mid E_1 \cap E_2)$.



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$\mathcal{Q}$. Find $\Pr(S \mid E_1 \cap E_2)$.

$\mathcal{A}$. By Bayes' theorem, $\Pr(S \mid E_1 \cap E_2) = \frac{\Pr(E_1 \cap E_2 \mid S)\Pr(S)}{\Pr(E_1 \cap E_2)}$.

$\mathcal{A}$. By conditional independence, $\Pr(E_1 \cap E_2 \mid S) = \Pr(E_1 \mid S)\Pr(E_2 \mid S)$ and similarly for $\overline{S}$.

$\mathcal{A}$. Since the prior probabilities are equal,

$$\Pr(S \mid E_1 \cap E_2) = \frac{\Pr(E_1 \mid S)\Pr(E_2 \mid S)}{\Pr(E_1 \mid S)\Pr(E_2 \mid S) + \Pr(E_1 \mid \overline{S})\Pr(E_2 \mid \overline{S})} = \frac{p(w_1)p(w_2)}{p(w_1)p(w_2) + q(w_1)q(w_2)}$$

Bayesian Spam Filter: Multiple Words

Assume that the words “stock” and “undervalued” occur independently, and $p(S) = p(\overline{S}) = 1/2$.

- Among 2000 spam messages, “stock” occurs 400 times: $p(w_1) = \frac{400}{2000} = 0.2$.
- Among 1000 non-spam messages, “stock” occurs 60 times: $q(w_1) = \frac{60}{1000} = 0.06$.
- For “undervalued”, $p(w_2) = \frac{200}{2000} = 0.1, \qquad q(w_2) = \frac{25}{1000} = 0.025$.

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Using independence:

$$r(w_1, w_2) = \frac{p(w_1)p(w_2)}{p(w_1)p(w_2) + q(w_1)q(w_2)} = \frac{(0.2)(0.1)}{(0.2)(0.1) + (0.06)(0.025)} = \frac{0.02}{0.0215} \approx 0.9302$$

$$0.9302 > 0.9 \implies \text{classify as spam.}$$

Expected Value and Variance

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Definition

The **expected value** (mean) of a random variable X on sample space S is:

$$E(X) = \sum_{s \in S} \Pr(s) X(s)$$



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Alternative Definition

If X is a random variable and $\Pr(X = r)$ is the probability that $X = r$, then:

$$E(X) = \sum_{r \in X(S)} \Pr(X = r) \cdot r$$



Example

Example: Suppose in rolling a die, the random variable X denotes the outcome. X takes values $1, 2, 3, 4, 5, 6$ each with probability $1/6$.

$$E(X) = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = \frac{7}{2}$$

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Example: A ticket costs nothing and pays

- Rs. 100 with probability 0.10,
- Rs. 20 with probability 0.30,
- Rs. 0 with probability 0.60.

The expected winning per ticket: $E(X) = 100(0.10) + 20(0.30) + 0(0.60) = 10 + 6 = 16$. Thus, over many plays, the **average winning per ticket is Rs. 16**.



Interpreting Expected Value

The expected value is a **probability-weighted average** of the possible values of a random variable.

$$E(X) = \sum_{x} x \Pr(X = x).$$

It does **not** necessarily mean the value that we will actually observe.



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The expected value is a **probability-weighted average** of the possible values of a random variable.

$$E(X) = \sum_{x} x \Pr(X = x).$$

It does **not** necessarily mean the value that we will actually observe. For example, when rolling a fair die,

$$E(X) = \frac{7}{2} = 3.5.$$

But a single roll can never produce 3.5. Instead, the expected value describes the **long-run average** over many repetitions.





Thank You for your attention!

Any questions?